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INTERNSHIP REPORT

on

Attention-Guided Deep Learning Framework for Automated Brain Tumor Segmentation in FLAIR MRI

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Certificate

This is to certify that the project report titled “Attention-Guided Deep Learning Framework for Automated Brain Tumor Segmentation in FLAIR MRI” is a record of the work carried out by Noor ul Huda during the AICTE Samarthan Internship at Indian Institute of Information Technology, Design and Manufacturing (IIITDM) Kancheepuram.

The work presented in this report includes the study, implementation, experimentation, and evaluation of deep learning models for automatic brain tumor segmentation from magnetic resonance imaging data. The implementation uses the BraTS 2020 FLAIR MRI modality and corresponding ground-truth masks for supervised binary whole-tumor segmentation.

The report has been prepared in a formal academic style and includes the problem background, internship objectives, literature review, research gap, methodology, model architecture diagrams, experimental setup, evaluation metrics, results, conclusion, and future scope of the project.

Supervisor / Mentor Signature



DR. BHUKYA KRISHNA PRIYA

Declaration

I hereby declare that the project report titled “Attention-Guided Deep Learning Framework for Automated Brain Tumor Segmentation in FLAIR MRI” is based on the work carried out as part of my internship project. The report has been prepared using the implementation, experiments, observations, and results obtained during the project work.

The project is based on supervised deep learning-based medical image segmentation. It uses labelled FLAIR MRI images and corresponding ground-truth tumor masks from the BraTS 2020 dataset. The work does not claim semi-supervised learning as part of the implemented pipeline.

All external sources of information, including research papers, datasets, software libraries and tools, have been acknowledged in the relevant sections and in the reference list.

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I also acknowledge the contribution of the BraTS research community, whose benchmark dataset and related publications made it possible to study brain tumor segmentation in a structured way. This project improved my understanding of convolutional neural networks, U-Net architectures, attention mechanisms, loss functions, segmentation metrics, and the practical challenges of deep learning models under constrained computational resources.

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Abstract

Brain tumor segmentation is an important task in medical image analysis because it helps identify the location and extent of tumor regions in MRI scans. Manual segmentation is time-consuming, depends on expert interpretation, and may vary between delineators. This project focuses on automatic binary whole-tumor segmentation using the BraTS 2020 FLAIR MRI data.

The work focuses on the FLAIR modality because FLAIR images provide useful contrast for whole-tumor abnormalities, especially edema and hyper-intense tissue. The original BraTS tumor labels were converted into a binary mask in which all tumor labels were treated as foreground and the remaining region was treated as background. The project follows a supervised learning approach using labelled FLAIR MRI slices and their corresponding ground-truth masks.

The implemented pipeline includes NIfTI image loading, FLAIR extraction, intensity normalization, axial slice preparation, binary mask generation, patient-level train-validation-test splitting, model training, threshold tuning, connected-component post-processing, and patient-level evaluation. The models considered in the project include an improved FLAIR-only 2D U-Net, a self-attention U-Net, a cross-attention-gate U-Net, and an attention ensemble.

The final selected attention ensemble achieved an average validation Dice score of 0.8776 and an average test Dice score of 0.8667. The test IoU, sensitivity, precision, and specificity were 0.7741, 0.8678, 0.8824, and 0.9988 respectively. The results show that meaningful whole-tumor segmentation can be achieved using FLAIR-only MRI, and the approach can be extended further through multimodal or 3D modelling in future work.

Chapter 1

Introduction

1.1 Background of Brain Tumor Segmentation

Medical image segmentation is the process of dividing a medical image into meaningful regions so that a particular anatomical structure or abnormal region can be identified, measured, and studied. In brain tumor analysis, segmentation means separating tumor tissue from surrounding healthy brain tissue in MRI scans. This task is important because a properly segmented tumor region can support tumor-size estimation, location analysis, treatment planning, follow-up assessment, and comparison of disease progression across scans. The need for standardized brain tumor segmentation datasets and evaluation protocols has been emphasized by the BraTS benchmark work [1], and curated glioma MRI collections with expert labels have further supported supervised segmentation research [2].

Manual tumor delineation is still valuable because it is performed by trained experts, but it is time-consuming and can vary between observers. Brain tumors also differ greatly in shape, size, intensity, boundary clarity, and position. These variations make automatic segmentation difficult, especially when the model has to identify small or diffuse tumor regions. The automatic segmentation problem therefore requires methods that can learn both local image details and wider contextual information from MRI scans [3,4,5].

1.2 MRI Modalities and BraTS Dataset

MRI is widely used for brain tumor analysis because it provides high soft-tissue contrast without ionizing radiation. In the BraTS dataset, the common modalities include T1, contrast-enhanced T1 (T1ce), T2, and FLAIR, and each modality highlights different tissue characteristics [1,18]. Multimodal MRI is useful because combining modalities can provide richer information about tumor boundaries, edema, enhancing regions, and necrotic areas. However, multimodal systems are more complex because they require all modalities to be available, aligned, preprocessed, and loaded together.

This project uses the BraTS 2020 dataset and focuses specifically on the FLAIR modality [18]. FLAIR is useful for whole-tumor segmentation because edema and hyperintense tumor-related abnormalities are often visible in this sequence. At the same time, using only FLAIR makes the task more challenging because complementary information from T1, T1ce, and T2 is not used. For this reason, the project studies a practical supervised FLAIR-only pipeline rather than a full multimodal pipeline.

1.3 Deep Learning for Medical Image Segmentation

Deep learning methods are suitable for image segmentation because they can learn useful visual features directly from image data. Early CNN-based methods showed that deep convolutional networks can learn tumor-related patterns from MRI images and improve automatic brain tumor segmentation [3,4]. Multi-scale 3D CNN approaches also showed that spatial context and post-processing are important for accurate brain lesion segmentation [5].

U-Net is one of the most influential architectures for biomedical image segmentation because it combines an encoder-decoder structure with skip connections [6]. The encoder captures high-level contextual features, while the decoder reconstructs the segmentation output at the original spatial resolution. Volumetric methods such as 3D U-Net and V-Net extended this idea to 3D medical images and showed the importance of volumetric context and Dice-based optimization [7,8]. However, these 3D methods require greater GPU memory and longer training time, which makes a 2D FLAIR-only approach more suitable for the computational scope of this internship project.

1.4 Pipeline Design, Attention, and Class Imbalance

Brain tumor segmentation performance depends not only on the architecture but also on the complete training and evaluation pipeline. BraTS-specific approaches, including cascaded CNN designs, autoencoder-regularized models, and nnU-Net-style configurations, show that preprocessing, training strategy, validation procedure, and post-processing all contribute to final segmentation performance [9,10,11,12]. This supports the pipeline-based design followed in the this project.

Attention mechanisms are useful in medical segmentation because they can help the network focus on informative regions and reduce the effect of irrelevant background features. Attention U-Net introduced attention gates for feature refinement, while UNet++ explored improved skip-pathway design for reducing the semantic gap between encoder and decoder features [13,14]. These ideas are relevant to this project because the self-attention and cross-attention-gate models are used to study whether attention-based feature refinement improves FLAIR-only tumor

segmentation.

Class imbalance is another important issue in tumor segmentation. In MRI slices, the background region is usually much larger than the tumor region. Dice-based loss, generalized Dice loss, Tversky loss, and focal loss have been studied to address imbalance and improve learning from difficult foreground regions [15,16,17]. In this project, this background motivates the use of overlap-based and classification-based evaluation metrics, including Dice score, IoU, sensitivity, precision, and specificity.

1.5 Focus of This Work

This project develops a supervised FLAIR-only brain tumor segmentation pipeline using U-Net and attention-based U-Net models. The original BraTS tumor labels are converted into a binary whole-tumor mask, and the model is trained to separate tumor regions from background. The implementation uses Python-based deep learning tools, including PyTorch for model development [19] and NiBabel for handling NIfTI medical image files [20].

The main purpose of this work is to build a clear and reproducible internship-level deep learning pipeline for medical image segmentation. The project compares an improved 2D U-Net baseline with self-attention and cross-attention-gate U-Net variants, applies validation-based threshold tuning and connected-component post-processing, and evaluates the final predictions at patient level using Dice score, IoU, sensitivity, precision, and specificity.

1.6 Organization of the Report

The remaining chapters are arranged in a systematic order. The Introduction chapter includes the project background, focus of the work, motivation, formal problem statement, internship objectives, expected output, scope, and dataset description. The Literature Review chapter discusses previous work and includes the research gap addressed by this project. The Methodology chapter explains the implementation steps, while the Architecture and Implementation chapter describes the model structure. The Experimental Setup, Evaluation Metrics, and Results and Discussion chapters present the configuration, metrics, results, and interpretation. Finally, the Conclusion chapter summarizes the work and future scope.

1.7 Motivation

Accurate brain tumor segmentation of MRI scans is critical in medical image analysis to provide reliable diagnosis, guidance for treatment planning, and monitoring. The boundaries of the tumor are currently marked manually by a clinician or medical expert in practice. It is time consuming and needs domain expertise and it can have inter-rater variance. Auto segmentation is therefore essential to ensure it can be used quickly and efficiently.

Deep learning techniques for image segmentation are effective in medical image analysis as they can learn informative image features from MRI images. Among different types of MRI scans, FLAIR can be particularly useful in visualizing lesions or abnormal tissues, including brain tumors and oedema. However, using the FLAIR modality is difficult as it yields less information as a standalone scan than as a part of a multi-modal scan. Tumor structures usually possess irregular shapes and blurry borders with large variability between different patients.

In this project, motivated to achieve a feasible FLAIR-only brain tumor segmentation, an Attention U-Net based approach was developed. An Attention mechanism is incorporated to assist the network in learning better representations of tumors, focusing on regions important for identifying tumors while suppressing areas less important or irrelevant. The overall objective is to achieve effective tumor segmentation in FLAIR images and measure performance using established evaluation metrics such as Dice coefficient, intersection-over-union (IoU), sensitivity, and precision.

1.8 Problem Statement and Objectives

1.8.1 Problem Context

Brain tumor segmentation from MRI scans is an important task because the tumor region must be separated from normal brain tissue before its size, shape, and location can be studied. In manual segmentation, the tumor boundary is marked by a medical expert. Although expert annotation is reliable, it is time-consuming and may vary between observers, especially when tumor boundaries are unclear or irregular.

The BraTS 2020 dataset provides MRI volumes and expert segmentation masks for brain tumor segmentation research [18]. Since this project is an internship-level implementation with constrained training time and GPU resources, the work uses only the FLAIR modality and formulates the task as binary whole-tumor segmentation. This makes the problem focused, reproducible, and suitable for studying the complete deep learning pipeline from dataset preparation to patient-level evaluation.

1.8.2 Formal Problem Statement

The problem considered in this project is the automatic segmentation of binary whole-tumor regions from FLAIR MRI scans using supervised deep learning. Given a 2D axial FLAIR slice extracted from the BraTS 2020 dataset, the model must predict a binary mask in which tumor pixels are marked as foreground and non-tumor pixels are marked as background.

The model is trained using labelled FLAIR image-mask pairs. The original BraTS tumor labels are converted into a binary whole-tumor mask by merging all non-background tumor labels into one foreground class. The predicted mask is compared with the ground-truth mask using overlap-based and classification-based metrics such as Dice score, IoU, sensitivity, precision, and specificity.

1.8.3 Internship Objectives

The main objectives of the internship project are:

1. To study the BraTS 2020 brain tumor segmentation dataset and understand the role of the FLAIR MRI modality in whole-tumor visualization.
2. To convert the original BraTS tumor labels into a binary whole-tumor segmentation mask suitable for supervised learning.
3. To prepare a 2D slice-based segmentation dataset from FLAIR MRI volumes using intensity normalization and patient-level splitting.
4. To implement an improved FLAIR-only 2D U-Net baseline for binary whole-tumor segmentation.
5. To implement self-attention and cross-attention-gate U-Net variants for experimental comparison with the improved baseline.
6. To apply validation-based threshold tuning and connected-component post-processing to refine predicted masks.
7. To evaluate model performance using Dice score, IoU, sensitivity, precision, and specificity at patient level.
8. To compare the project results with related segmentation approaches and identify future improvements for the pipeline.

1.8.4 Expected Output

The expected output of the project is a predicted binary tumor mask for each FLAIR MRI slice or reconstructed patient-level case. The mask should highlight the tumor region and suppress the background. The final output is not intended for direct clinical diagnosis; it is an academic segmentation result that demonstrates how a supervised U-Net-based pipeline can be implemented and evaluated.

1.8.5 Scope of the Project

The scope of this project is restricted to FLAIR-only binary whole-tumor segmentation. The project does not perform tumor grading, survival prediction, clinical diagnosis, or multi-class tumor sub-region segmentation. It focuses on dataset preparation, preprocessing, model development, post-processing, evaluation, and result interpretation.

1.9 Dataset Description

1.9.1 Dataset Source

The dataset used in this project is the BraTS 2020 brain tumor segmentation dataset [18]. BraTS provides expert-annotated glioma MRI scans and is one of the most widely used benchmark datasets for developing and evaluating brain tumor segmentation methods. The use of BraTS in this project ensures that the experiments are grounded in a standardized and academically accepted dataset.

1.9.2 MRI Modalities and Selected Input

Each BraTS case contains four MRI modalities: T1, T1ce, T2, and FLAIR. Although the full dataset is multimodal, this project deliberately uses only the FLAIR sequence. FLAIR images are especially useful for highlighting edema and abnormal hyperintense tumor-related tissue, which makes them suitable for whole-tumor segmentation. Restricting the project to FLAIR-only input also supports the internship goal of building a simpler and more reproducible supervised segmentation pipeline.

1.9.3 Segmentation Target

The original BraTS labels contain multiple tumor-related classes. In this work, labels 1, 2, and 4 were merged into a single foreground class in order to create a binary whole-tumor mask. All remaining pixels were treated as background. This conversion simplified the task from multi-class sub-region segmentation to binary whole-tumor segmentation while preserving the main objective of detecting the overall tumor extent.

1.9.4 Relevance to This Work

The FLAIR slice acts as the model input, while the converted binary tumor mask acts as the ground-truth target during supervised learning. Therefore, the dataset supports the full pipeline used in this project, including preprocessing, model training, validation, testing, and patient-wise evaluation.

Chapter 2

Literature Review

This literature review focuses on supervised FLAIR-only brain tumor segmentation using U-Net and attention-based U-Net models. The reviewed work is grouped into major areas relevant to this project, including the BraTS benchmark and annotated glioma MRI datasets, early CNN-based brain tumor segmentation, U-Net and volumetric medical segmentation, attention mechanisms, BraTS-specific segmentation methods, and loss functions for class imbalance. These studies provide the background required to understand the dataset, model selection, training strategy, and evaluation metrics used in this work.

2.1 BraTS Benchmark and Annotated Glioma MRI Data

Menze et al. Proposed the Multimodal Brain Tumor Image Segmentation Benchmark, more informally known as BraTS [1]. The major missing piece to be solved by this paper was that there was no available standardized benchmark which would have allowed evaluation of brain tumor segmentation algorithms in the field. They made available multimodal MRIs with expert annotations and a standardized metric for evaluating the results. The project at hand relates directly as it employs data from BraTS and uses overlap-based measures to assess the accuracy of segmentation results.

The glioma MRI collection was extended with expert segmentation labels and radiomic features in [2] by Bakas et al. The problem that they aim to solve was the scarcity of curated and reproducible data with annotation for glioma researchers to learn and to train their algorithms. Labelled MRI data is necessary for the field since they are the standard type of input used for supervised learning of segmentation systems. As stated previously, supervised learning forms the cornerstone of this project.

2.2 Early CNN-Based Brain Tumor Segmentation

Deep neural networks were introduced for brain tumor segmentation by Havaei et al [3]. The researchers observed that hand-crafted features could not account for the complex appearance of the tumor in the MRI images and that a deep convolutional network could automatically extract feature information directly from the image to improve the accuracy of automatic segmentation of the tumor in the brain. This motivates deep learning as an excellent method for this project.

Using small kernel CNNs for brain tumor segmentation in MRI images was also described by Pereira et al [4]. Their work addressed the issues with tumor variation in size, shape, location, and intensity. It also used preprocessing techniques, such as augmentation of the training data to create a more robust network. The significance for this project is that supervised CNNs can learn tumor features on similar data to what we expect to receive.

Kamnitsas et al [5] proposed a 3D multi-scale CNN architecture which is closely related to DeepMedic for segmentation of brain lesions. This work highlights the issue that while we are using a 2D segmentation model for computational reasons, context is extremely important and as is post-processing.

2.3 U-Net and Volumetric Medical Segmentation

Ronneberger et al. Published U-Net for segmentation in biomedical imaging [6]. U-Net was developed to overcome the challenge of extracting precise segmentations from very limited labeled biomedical images. Its encoder-decoder architecture and skip connections allow the network to benefit from the combining of context and localization information. The architecture in this project relies on U-Net.

Cicek et al. Generalized U-Net for 3D volumetric segmentation [7]. This is essential, since medical image data from an MRI are naturally volumetric. However, 3D segmentation also significantly increases demand on computational hardware, as it requires more GPU RAM and can take longer to train. Due to these restrictions in terms of computational requirements in the context of an internship, this work follows a 2D slice-based methodology. Milletari et al. Developed V-Net and presented a Dice-based objective for optimizing volumetric medical image segmentation [8].

A Dice-based optimization is crucial, as tumor segmentation often is dominated by a significant class imbalance, with the tumor area being small in proportion to the image region.

This provides another rationale for a Dice-based approach in both training and evaluating the segmentations generated in this project.

2.4 BraTS-Specific Segmentation Pipelines and Optimization

Wang et al. proposed a cascaded anisotropic convolutional network for automatic brain tumor segmentation [9]. Their work is relevant because it shows how segmentation can be organized in stages and how post-processing can help reduce false predictions. Myronenko used an encoder-decoder design with autoencoder regularization for BraTS segmentation [10]. This supports the idea that encoder-decoder models can be strengthened through regularization and careful architectural design.

Isensee et al. introduced nnU-Net as a self-configuring method for biomedical image segmentation [11]. The importance of nnU-Net for this project is that it shows that segmentation performance depends not only on the architecture, but also on preprocessing, training configuration, validation strategy, and post-processing. The related No New-Net work also demonstrated that strong BraTS performance can be achieved through systematic pipeline design rather than only by adding new network blocks [12]. These studies support the pipeline-based approach followed in this internship project.

2.5 Attention and Skip-Connection Refinement

Oktay et al. Proposed Attention U-Net with attention gates [13]. The problem to be solved was that attention gates do not carry the irrelevant background information with it, while the standard skip connections carry everything from encoder features into the decoder. It filters away uninteresting features, highlights structures of interest.

The attention-gate experiment is the direct motivation for the cross-attention-gate U-Net experiment in this paper.

Zhou et al. Designed UNet++ [14] that utilizes nested skip pathways in order to minimize the semantic differences between features in the encoder and decoder branches. In the scope of this paper, its motivation for nested skip connections highlight the relevance of feature merging in segmentation architectures. Both attention gate and nested skip pathways enhance the information passed from encoder to decoder features. Attention gates are helpful in medical image segmentation since the structure of interest could take up a tiny region of a big background, thus attention can force the network to be attentive towards bright hyperintense regions when segmenting FLAIR-only brain tumors.

Nonetheless, the findings here illustrate that it's not always optimal and need complementation by appropriate preprocessing, sampling and postprocessing procedures.

2.6 Loss Functions for Imbalanced Segmentation

Sudre et al. Investigated Generalized Dice loss in unbalanced segmentations [15]. This approach helps reduce the problem that the background may have far more pixels than the foreground in such unbalanced scenarios, which is exactly the situation of brain MRI segmentation, where the percentage of tumor is quite low. Salehi et al. Then proposed Tversky loss which could be adjusted to balance the false positives and false negatives [16]. Given that just high precision is not enough, and some parts of tumor can be easily overlooked with high precision but low sensitivity as also observed in our result analysis in the analysis of difficult cases, this is indeed a useful loss for tumor segmentation.

Lin et al. Also proposed focal loss for dense predictions [17], though it was initially designed for object detection, the idea is also applicable for segmentation under extreme class imbalance: it essentially down-weights the loss from easily predicted examples (the easy backgrounds in segmentation) and focuses the learning on the hard foreground samples.

2.7 Summary of Literature Review

The following table summarizes only the papers and research works that are discussed in the Literature Review section. Along with the method and evaluation metrics, the table also reports the actual quantitative scores whenever the cited paper provides them. For dataset papers or loss-function papers that do not present one direct BraTS segmentation score, the result column states the reported contribution clearly instead of inventing a numerical value.

Table 2.1: Summary of literature reviewed with metrics and reported scores

Title / Work	Author(s)	Method	Metrics	Reported Scores / Results
Multimodal Brain Tumor Image Segmentation Benchmark (BraTS) [1]	Menze et al.	Standardized multimodal MRI benchmark with expert tumor annotations and common evaluation protocol.	Dice, sensitivity, specificity, Hausdorff distance.	Benchmark paper; no single model score. Establishes the BraTS evaluation protocol used by later studies.
Glioma MRI collections with expert segmentation labels [2]	Bakas et al.	Curated glioma MRI collections with expert labels and radiomic features.	Dataset quality, annotation consistency, reproducibility.	Data-resource paper; no proposed model score. Supports supervised training with labelled glioma MRI data.
Brain tumor segmentation with deep neural networks [3]	Havaei et al.	Two-pathway CNN and cascaded CNN for multimodal BraTS tumor segmentation.	Dice, sensitivity, specificity.	BRATS 2013 test Input-CascadeCNN*: Dice complete/core/enhancing = 0.88/0.79/0.73; sensitivity = 0.87/0.79/0.80; specificity = 0.89/0.79/0.68.
Brain tumor segmentation using CNNs in MRI images [4]	Pereira et al.	Small-kernel CNN with intensity normalization, augmentation, and post-processing.	Dice similarity coefficient.	BRATS 2013: Dice complete/core/enhancing = 0.88/0.83/0.77. BRATS 2015: Dice = 0.78/0.65/0.75 for complete/core/enhancing regions.
Efficient multi-scale 3D CNN / DeepMedic [5]	Kamnitsas et al.	3D multi-scale CNN with dense training and CRF post-processing.	DSC, precision, sensitivity.	BRATS 2015 test Ensemble+CRF: DSC whole/core/enhancing = 84.9/66.7/63.4%; precision = 85.3/86.1/63.4%; sensitivity = 87.7/60.0/67.4%.

Title / Work	Author(s)	Method	Metrics	Reported Scores / Results
U-Net [6]	Ronneberger et al.	Encoder-decoder CNN with skip connections for biomedical segmentation.	Warping error, Rand error, IoU/Jaccard in challenge tasks.	ISBI EM segmentation: warping error = 0.0003529 and Rand error = 0.0382; U-Net also won ISBI cell tracking challenge categories.
3D U-Net [7]	Cicek et al.	Extended U-Net to 3D volumetric segmentation with sparse annotation.	IoU.	Semi-automated Xenopus kidney segmentation: average IoU = 0.863 with batch normalization; fully automated setting average IoU = 0.704.
V-Net [8]	Milletari et al.	Volumetric fully convolutional network trained with Dice-based objective.	Average Dice, average Hausdorff distance.	PROMISE 2012 prostate MRI: V-Net + Dice loss reported average Dice = 0.869 ± 0.033 and average Hausdorff distance = 5.71 ± 1.20 mm.
Cascaded anisotropic CNN for brain tumor segmentation [9]	Wang et al.	Cascaded anisotropic and dilated CNNs with multi-view fusion for BraTS subregions.	Dice.	BraTS 2017 validation Dice ET/WT/TC = 0.7859/0.9050/0.8378; testing Dice ET/WT/TC = 0.7831/0.8739/0.7748.
3D MRI brain tumor segmentation with autoencoder regularization [10]	Myronenko	Encoder-decoder segmentation network with variational autoencoder regularization.	Dice, Hausdorff distance.	BraTS 2018 testing average Dice ET/WT/TC = 0.7664/0.8839/0.8154; won first place in BraTS 2018.
nnU-Net [11]	Isensee et al.	Self-configuring biomedical segmentation framework adapting preprocessing, architecture and training settings.	Dice and task-specific metrics across datasets.	General framework paper; no single BraTS score is the main result. Shows automatic configuration improves many segmentation tasks.
No New-Net [12]	Isensee et al.	Optimized BraTS pipeline using U-Net-style architecture, region-based training and post-processing.	Dice, Hausdorff distance.	BraTS 2018 validation baseline Dice ET/WT/TC = 79.59/90.80/84.32%; final pipeline improved through region-based training and post-processing.
Attention U-Net [13]	Oktay et al.	Attention gates added to U-Net skip connections for pancreas segmentation.	Dice score, surface-to-surface distance.	Proposed Attention U-Net reported mean Dice around 85.99% for pancreas segmentation and motivated attention-gate feature filtering.
UNet++ [14]	Zhou et al.	Nested and dense skip pathways with deep supervision.	IoU, Dice.	UNet++ with deep supervision reported average IoU gains of 3.9 points over U-Net and 3.4 points over wide U-Net across medical segmentation tasks.
Generalized Dice loss [15]	Sudre et al.	Generalized Dice loss for highly imbalanced segmentation tasks.	Generalized Dice, Dice-style overlap measures.	Loss-function study; no single BraTS tumor Dice score. It showed that generalized Dice is robust for class-imbalanced segmentation.
Tversky loss [16]	Salehi et al.	Tversky loss to balance false positives and false negatives in medical segmentation.	Dice/Tversky-style overlap, sensitivity-precision balance.	Loss-function study; no direct BraTS score. Improves FP/FN control and is relevant for tumor sensitivity.
Focal loss [17]	Lin et al.	Focal loss to reduce the effect of easy examples and focus on hard examples.	Average precision in dense detection.	Original detection paper; no tumor Dice score. Shows hard-example weighting improves dense prediction AP.

2.8 Research Gap

Existing research papers have reported strong results for brain tumor segmentation, but many of them rely on multimodal MRI inputs, 3D architectures, or more computationally intensive training strategies [5,7,9,10,11]. In contrast, comparatively fewer studies focus on a simplified and reproducible FLAIR-only setting for binary whole-tumor segmentation. This creates a practical research gap for internship-level work where computational resources and project duration are constrained.

2.8.1 Gap Identified from Previous Studies

Most earlier studies either use all four BraTS modalities, focus on volumetric 3D segmentation, or target more complex multi-class tumor sub-region segmentation [1,5,7,10,11]. Such approaches are effective, but they are not always easy to reproduce in a short internship setting because they require higher GPU memory, longer training time, and more complex preprocessing.

Several attention-based segmentation studies show that attention mechanisms can improve feature selection [13,14], but their effectiveness in a supervised FLAIR-only binary segmentation setting is not always examined in a simple and comparable pipeline. Similarly, loss-function studies address class imbalance [15,16,17], but there remains a need to study how far a practical FLAIR-only U-Net pipeline can be improved through preprocessing, attention modules, threshold tuning, and post-processing without moving to much heavier architectures.

2.8.2 Gap Addressed by This Project

This project addresses this gap by developing a supervised brain tumor segmentation pipeline that:

- uses only the FLAIR modality from BraTS 2020,

- converts the original labels into a binary whole-tumor mask for a simplified but meaningful segmentation task,
- compares an improved 2D U-Net baseline with self-attention and cross-attention-gate U-Net variants,
- applies validation-based threshold tuning and connected-component post-processing, and
- evaluates the models using patient-level Dice, IoU, sensitivity, precision, and specificity.

Thus, the contribution of this work is not to claim state-of-the-art performance, but to study how an efficient, understandable, and reproducible FLAIR-only segmentation pipeline can be built and evaluated within the scope of an internship project.

Chapter 3

Methodology

3.1 Dataset Selection and Task Definition

The project uses the BraTS 2020 dataset [18], which provides pre-processed multimodal MRI volumes and corresponding expert annotations for glioma segmentation. Although BraTS includes multiple MRI modalities, this work deliberately restricts the input to the FLAIR sequence. This choice was made to study how far a single-modality pipeline can perform in a supervised setting while remaining computationally manageable during the internship.

The original BraTS segmentation labels were converted into a binary whole-tumor mask. In this binary setting, all tumor-related labels were merged into a single foreground class and all remaining pixels were treated as background. This simplified the task from multi-class tumor sub-region segmentation to binary whole-tumor segmentation while still retaining clinical relevance.

3.2 Preprocessing and Slice Preparation

The MRI data were handled in NIfTI format using the NiBabel-based medical image loading workflow [20]. For each patient case, the FLAIR volume and its corresponding segmentation mask were loaded. The FLAIR intensities were normalized in order to reduce intensity variation across patient scans. The 3D volumes were then converted into axial 2D slices, because the implemented pipeline follows a slice-based 2D training strategy.

Each axial slice was paired with the corresponding binary tumor mask. This produced image-mask pairs that could be directly used for supervised learning. The slice preparation stage also ensured that the model received spatially aligned input-target pairs during training and evaluation.

3.3 Patient-Level Splitting Strategy

To avoid data leakage, the dataset was split at the patient level rather than at the slice level. This means that slices from the same patient did not appear across both training and validation or testing sets. This step is important in medical image analysis, because slice-level splitting can lead to overly optimistic results when neighboring slices of the same patient are seen during both training and testing.

The training split was used for model optimization, the validation split was used for model selection and threshold tuning, and the test split was reserved only for final evaluation.

3.4 Model Development

The project began with an improved FLAIR-only 2D U-Net baseline and was then extended with attention-based variants. Two attention models were studied in detail: a self-attention U-Net and a cross-attention-gate U-Net. These models were designed to examine whether attention mechanisms could improve tumor localization and feature selection in a FLAIR-only setup.

The self-attention model incorporates an attention operation at the bottleneck layer so that the network can model long-range feature relationships. The cross-attention-gate model applies attention gates in the skip pathways so that more relevant encoder features are passed to the decoder. In addition, an ensemble prediction strategy was considered in order to combine the strengths of the attention-based models.

3.5 Validation, Post-Processing and Final Evaluation

The models were trained using labelled image-mask pairs and the best checkpoints were selected using validation Dice score. After training, threshold tuning was carried out on the validation predictions. Connected-component post-processing was then applied to suppress small isolated false-positive regions.

The final selected configuration used a probability threshold of 0.50 and a minimum connected component size of 250. The selected setting was then evaluated at patient level on the test set. Final evaluation was reported using Dice score, Intersection over Union (IoU), sensitivity, precision, and specificity.

3.6 Step-by-Step Methodology

1. Load BraTS 2020 MRI data and identify the FLAIR modality.
2. Convert the original BraTS labels into a binary whole-tumor mask.
3. Normalize FLAIR intensities and prepare axial 2D slices.
4. Split patients into training, validation, and testing groups.
5. Train the improved 2D U-Net, self-attention U-Net, and cross-attention-gate U-Net using labelled image-mask pairs.
6. Save the best checkpoints based on validation Dice score.
7. Tune threshold and connected-component post-processing on validation predictions.
8. Generate ensemble predictions from the attention-based models.
9. Evaluate the selected configuration on the test set at patient level.
10. Interpret both quantitative and qualitative results.

Chapter 4

Architecture and Implementation

4.1 Overall Project Architecture

The overall project architecture explains how the input MRI data moves through each stage of the proposed segmentation system. The pipeline begins with BraTS 2020 MRI volumes and ends with patient-level evaluation of predicted tumor masks. Each block in Figure 4.1 shows both what the stage does and how it contributes to the final segmentation output.

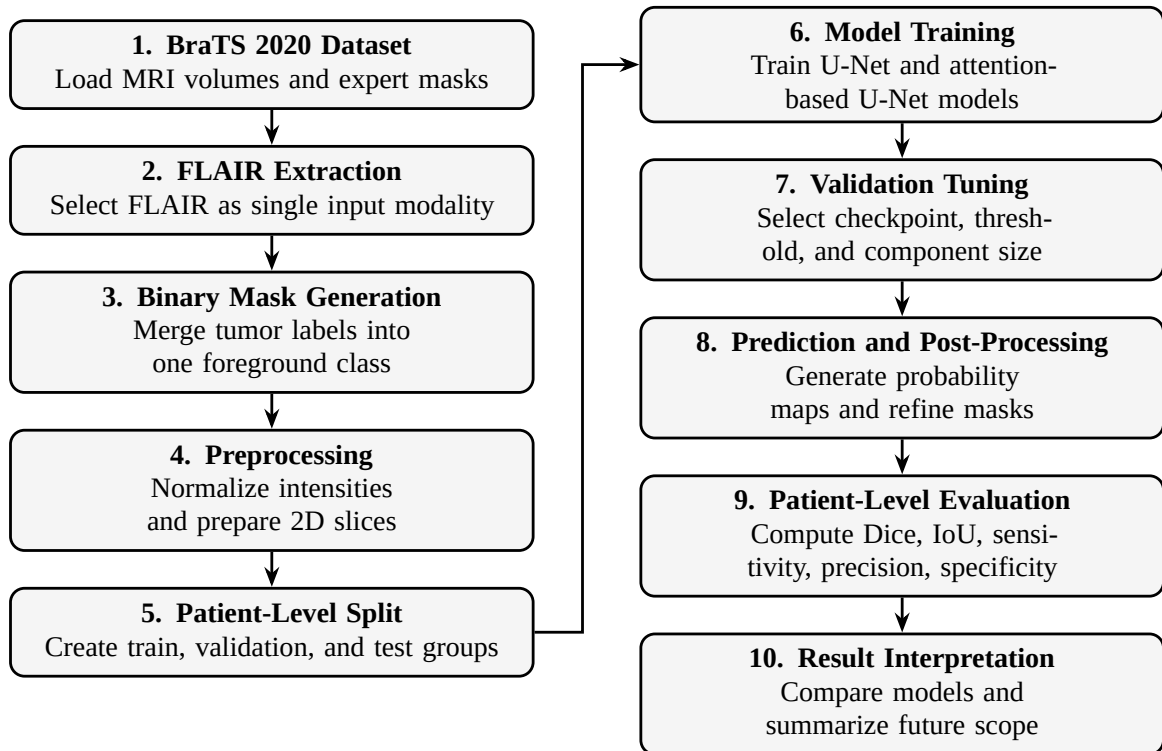


Figure 4.1: Step-wise block diagram showing the complete project working.

4.2 Improved 2D U-Net Baseline

The starting point of the implementation was an improved FLAIR-only 2D U-Net. U-Net consists of an encoder, a bottleneck, and a decoder. The encoder progressively extracts feature representations from the input image, while the decoder reconstructs a segmentation map at the original spatial resolution. Skip connections are used to transfer high-resolution spatial information from encoder layers to the decoder.

The improved baseline serves as an important reference model in the project. It helps determine whether later attention-based modifications produce meaningful improvement over a strong conventional U-Net pipeline.

4.3 Attention-Based U-Net Architecture

Attention-based extensions were introduced to investigate whether feature refinement could improve segmentation performance in the FLAIR-only setting. In the attention-based U-Net design, the core encoder-decoder structure remains the same, but the flow of information is refined either through attention at the bottleneck or through attention gates in skip connections.

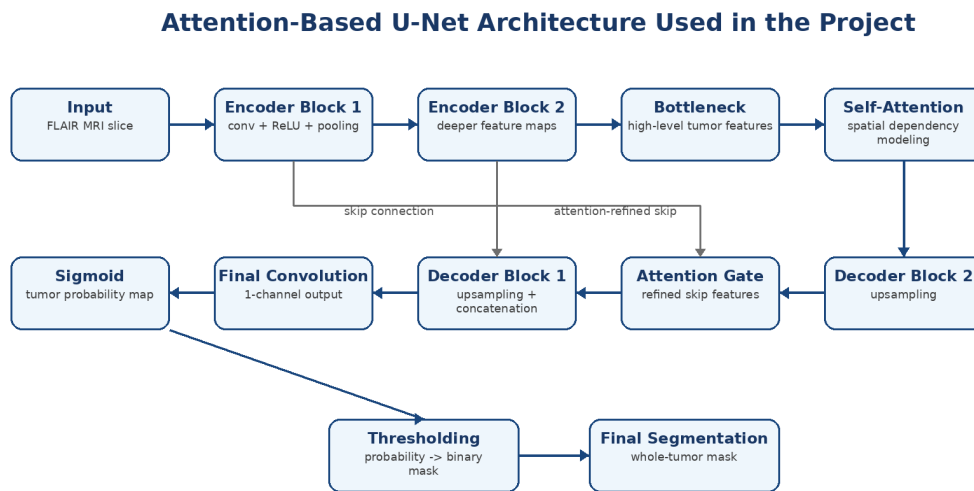


Figure 4.2: Attention-based U-Net architecture used in the project.

4.4 Self-Attention U-Net

The self-attention U-Net includes a self-attention block at the bottleneck layer. This allows the model to capture long-range spatial dependencies in the feature map. Such behavior is useful in

brain tumor segmentation because tumor regions may be spatially scattered, and a purely local convolution may not always capture their wider contextual relationship.

4.5 Cross-Attention-Gate U-Net

The cross-attention-gate U-Net places attention gates in the skip connections between encoder and decoder. These gates help the model filter irrelevant background information and pass more informative tumor-related features to the decoder. This is particularly useful in FLAIR images, where background tissue can be dominant and tumor regions may occupy only a relatively small area.

4.6 Ensemble Prediction Strategy

In the final stage, outputs from the attention-based models were combined to form an ensemble prediction. Probability maps from individual models were averaged, and the resulting fused prediction was converted into a binary mask using the selected threshold. This ensemble strategy was intended to reduce model-specific errors and produce more stable final segmentations.

Chapter 5

Experimental Setup

5.1 Implementation Environment

The experiments were implemented in Python using PyTorch [19] and were executed on Kaggle/Colab GPU resources. Medical image volumes were handled in NIfTI format using NiBabel [20]. The models were trained in a supervised manner using FLAIR-only input slices and their corresponding binary tumor masks. Validation monitoring was used during training, and post-processing was applied after threshold selection.

5.2 Training and Selection Strategy

The training process focused on comparing an improved FLAIR-only 2D U-Net baseline with two attention-based variants, namely the self-attention U-Net and the cross-attention-gate U-Net. During training, validation Dice score was tracked in order to identify the best checkpoint for each model. After training, validation predictions were used to tune the threshold and connected-component post-processing parameters. Finally, the selected configuration was evaluated on the test set at patient level.

5.3 Configuration Summary

The key settings used in the final experimental pipeline are summarized below.

Item	Description
Dataset	BraTS 2020
Input modality	FLAIR only
Segmentation target	Binary whole-tumor mask
Learning type	Supervised learning
Image representation	2D axial slices
Best selected model	Attention ensemble
Best threshold	0.50
Minimum component size	250
Evaluation level	Patient-level test evaluation
Metrics	Dice, IoU, sensitivity, precision, specificity

Chapter 6

Evaluation Metrics

Evaluation metrics were used to measure how closely the predicted tumor masks matched the ground-truth masks. Since brain tumor segmentation is an imbalanced task, a single metric is not sufficient. Most pixels in an MRI slice belong to the background, while tumor pixels occupy a smaller region. Therefore, this project uses both overlap-based metrics and classification-based metrics.

6.1 Pixel-Level Confusion Terms

The predicted binary mask and the ground-truth mask are compared pixel by pixel. The following terms are used for metric calculation:

Table 6.1: Confusion matrix terms used in segmentation evaluation

Term	Meaning in this project
True Positive (TP)	Tumor pixel correctly predicted as tumor.
False Positive (FP)	Background pixel incorrectly predicted as tumor.
False Negative (FN)	Tumor pixel incorrectly predicted as background.
True Negative (TN)	Background pixel correctly predicted as background.

6.2 Dice Score

Dice score is the main overlap metric used in this project. It measures the similarity between the predicted tumor mask and the ground-truth tumor mask. A higher Dice score means better overlap between prediction and annotation.

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (6.1)$$

Dice is important for medical image segmentation because it directly measures whether the

model has segmented the same region as the expert annotation. Dice-based optimization and overlap-based evaluation are commonly used in biomedical segmentation and brain tumor segmentation studies [8,15].

6.3 Intersection over Union

Intersection over Union, also called IoU or Jaccard index, measures the ratio of the common area between the prediction and ground truth to their combined area.

$$IoU = \frac{TP}{TP + FP + FN} \quad (6.2)$$

IoU is stricter than Dice because it compares the intersection directly with the full union of predicted and true tumor regions. It is useful for understanding how much extra or missing area exists in the predicted mask.

6.4 Sensitivity, Precision and Specificity

Sensitivity measures how much of the actual tumor region was detected. It is important because missed tumor pixels reduce the completeness of the segmentation.

$$Sensitivity = \frac{TP}{TP + FN} \quad (6.3)$$

Precision measures how much of the predicted tumor region was actually correct. A high precision indicates that the model is not producing many false tumor regions.

$$Precision = \frac{TP}{TP + FP} \quad (6.4)$$

Specificity measures how well the model identifies background pixels. In MRI segmentation, specificity is usually high because the background class is much larger than the tumor class.

$$Specificity = \frac{TN}{TN + FP} \quad (6.5)$$

6.5 Metric Summary

The metrics used in the project are summarized in Table 6.2. Dice and IoU focus on spatial overlap, sensitivity measures tumor detection, precision measures correctness of predicted tumor pixels, and specificity measures background identification.

Table 6.2: Evaluation metrics used in the project

Metric	Formula	Purpose
Dice score	$\frac{2TP}{2TP+FP+FN}$	Measures overlap between predicted and ground-truth tumor masks.
IoU	$\frac{TP}{TP+FP+FN}$	Measures intersection over union of predicted and actual tumor regions.
Sensitivity	$\frac{TP}{TP+FN}$	Measures how much of the actual tumor was detected.
Precision	$\frac{TP}{TP+FP}$	Measures how much of the predicted tumor was correct.
Specificity	$\frac{TN}{TN+FP}$	Measures correct identification of background pixels.

6.6 Patient-Level Evaluation

Final performance was interpreted at patient level rather than only at isolated slice level. This is important because a patient case contains multiple MRI slices, and the final segmentation quality should reflect the overall patient volume. Patient-level evaluation gives a more realistic view of model behavior and reduces the chance of over-interpreting individual slices.

Chapter 7

Results and Discussion

7.1 Model Validation Summary

Table 7.1: Model validation and test summary

Model	Dice	IoU	Sensitivity	Precision	Specificity
Improved FLAIR-only 2D U-Net	0.8619	0.7672	0.8583	0.8834	0.9986
Self-attention U-Net	0.8741	0.7836	0.8769	0.8828	0.9989
Cross-attention-gate U-Net	0.8758	0.7859	0.8801	0.8812	0.9988
Attention ensemble	0.8667	0.7741	0.8678	0.8824	0.9988

Plain U-Net, enhanced, worked as a powerful baseline. Self-attention and cross-attention-gate resulted in excellent validation accuracy and attention ensemble gave the highest chosen Dice on testing. As shown from this result, final accuracy depended on not only attention method, but also pre-processing, training stability, threshold tuning, and post-processing.



Figure 7.1: Model performance summary.

7.2 Final Selected Performance

Table 7.2: Final validation and test performance of the selected ensemble model

Metric	Validation	Test
Dice score	0.8776	0.8667
Intersection over Union	0.7884	0.7741
Sensitivity	0.8794	0.8678
Precision	0.8856	0.8824
Specificity	0.9991	0.9988

The final selected model was the attention ensemble. The model was trained with a threshold of 0.50 and a minimum connected-component size of 250. The validation Dice score for the model was 0.8776 and the final test Dice score was 0.8667.

Test precision was 0.8824 and test sensitivity was 0.8678.

Overall, the model predicts fairly well in most regions of the tumors, but some regions were still omitted for more challenging examples. This is expected behavior since we were using FLAIR only segmentation and a absence of other imaging modalities (T1, T1ce, and T2), which would have offered additional information about tumor border, contrast, etc.

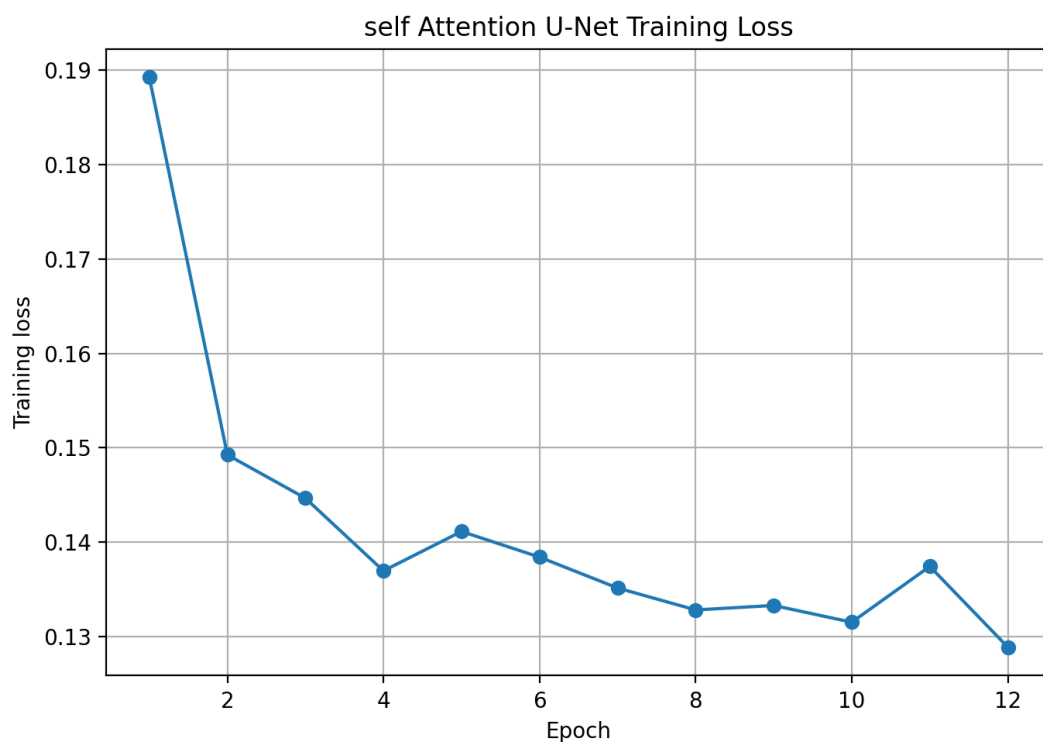


Figure 7.2: Self-attention U-Net training loss curve.

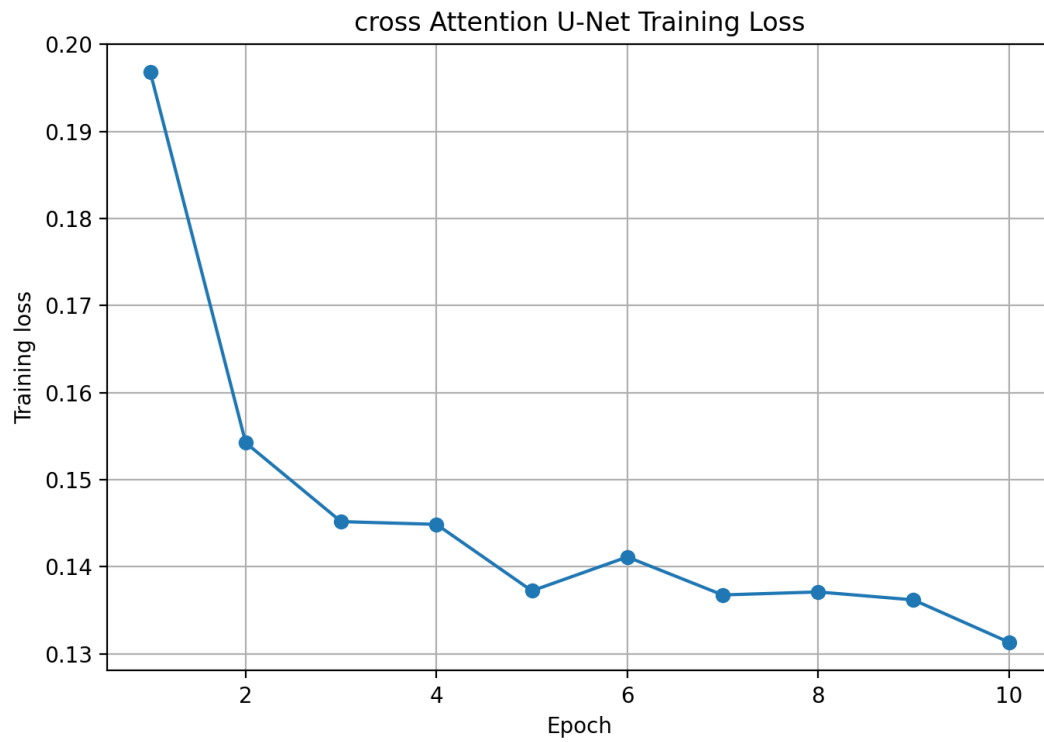


Figure 7.3: Cross-attention-gate U-Net training loss curve.

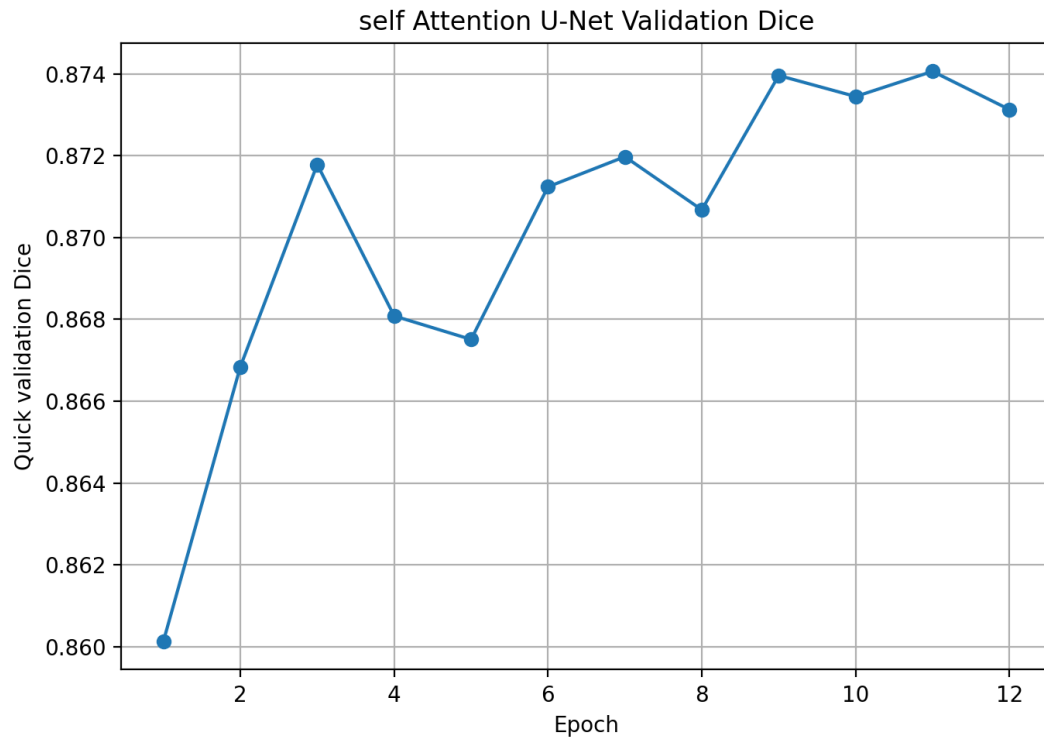


Figure 7.4: Self-attention U-Net validation Dice curve.

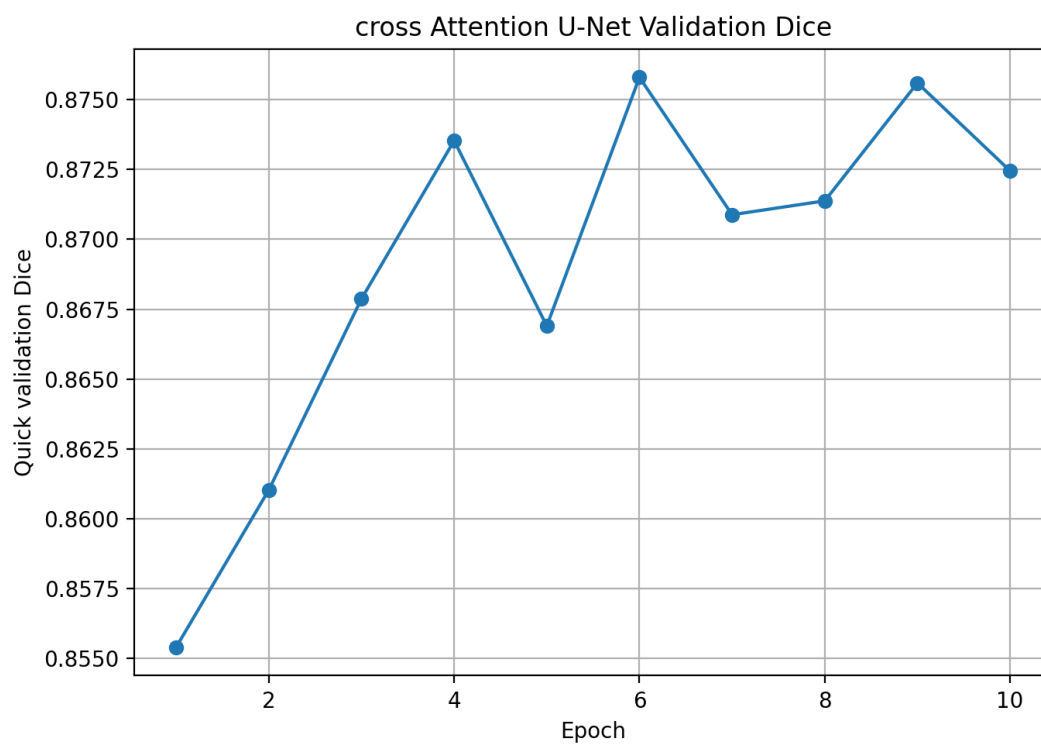


Figure 7.5: Cross-attention-gate U-Net validation Dice curve.

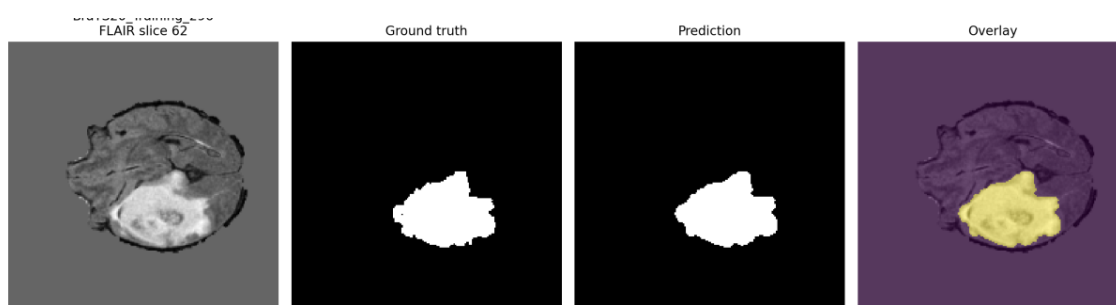


Figure 7.6: Example FLAIR slice, ground-truth mask and predicted mask.

7.3 Patient-Wise Results

Table 7.3: Top five patient-wise cases by Dice score

Case ID	Dice	IoU	Sensitivity	Precision
BraTS20_Training_296	0.9663	0.9347	0.9570	0.9758
BraTS20_Training_333	0.9609	0.9247	0.9523	0.9696
BraTS20_Training_362	0.9477	0.9007	0.9562	0.9394
BraTS20_Training_328	0.9439	0.8938	0.9417	0.9462
BraTS20_Training_259	0.9434	0.8928	0.9315	0.9556

Table 7.4: Difficult patient-wise cases by Dice score

Case ID	Dice	IoU	Sensitivity	Precision
BraTS20_Training_302	0.4992	0.3327	0.3331	0.9963
BraTS20_Training_048	0.6481	0.4794	0.4893	0.9594
BraTS20_Training_303	0.6511	0.4826	0.4991	0.9360
BraTS20_Training_136	0.7238	0.5671	0.6437	0.8266
BraTS20_Training_058	0.7326	0.5780	0.7476	0.7181

Top-performing cases demonstrate that the model can produce strong overlap when the tumor region is clearly visible in the FLAIR image. In more challenging cases, the lower Dice score mainly comes from missed tumor regions. In some cases, precision remains high while sensitivity is lower, which means that the predicted tumor pixels are mostly correct but the model does not cover the entire tumor area.

7.4 Comparison with Existing Research

The result of this project should be interpreted in relation to the scope of the work. Many earlier studies use multimodal MRI inputs, 3D architectures, cascaded designs, or highly optimized pipelines, whereas this project uses a simpler FLAIR-only 2D supervised setup. The following table compares this work with related research directions.

Table 7.5: Comparison of this project with related research approaches

Study / approach	Main focus	Relation to this project
BraTS benchmark and annotated datasets [1,2,18]	Standardized multimodal MRI data and expert masks	This project uses BraTS 2020 data and follows supervised evaluation using expert masks.
Early CNN methods [3,4]	Learning tumor features directly from MRI images	This project follows the same supervised learning idea but uses U-Net-based segmentation instead of earlier CNN designs.
DeepMedic and 3D models [5,7,8]	Multi-scale or volumetric medical image segmentation	These methods use richer spatial context, while this project uses a practical 2D FLAIR-only pipeline.
BraTS optimized pipelines [9,10,11,12]	Cascaded, regularized, or self-configuring segmentation pipelines	This project adopts the idea that preprocessing, validation, and post-processing are important parts of performance.
Attention and skip-refinement methods [13,14]	Feature selection and improved skip connections	This project implements self-attention and cross-attention-gate U-Net variants for comparison.
This project	FLAIR-only 2D U-Net, attention variants, and ensemble prediction	The final selected ensemble achieved Dice 0.8667, IoU 0.7741, sensitivity 0.8678, precision 0.8824, and specificity 0.9988 at patient level.

This comparison shows that this project is positioned as a focused and reproducible FLAIR-only segmentation pipeline. It does not attempt to replace more complex multimodal or 3D systems, but it demonstrates that a complete U-Net-based workflow can produce meaningful segmentation results under a simpler input setting.

7.5 Interpretation of Results

The overall results indicate that attention-based learning is helpful in the FLAIR-only setting, but it is not by itself the only reason for improved performance. The final outcome also depends on how carefully the pipeline is designed. In this project, dataset preparation, binary mask generation, patient-level splitting, threshold tuning, and connected-component post-processing all contributed to the final performance.

Another important observation is that high specificity was achieved across the models. This means that background tissue was generally classified correctly, while the main area for improvement is complete coverage of tumor pixels in challenging cases. Therefore, improving sensitivity remains an important direction for further work.

Chapter 8

Conclusion

8.1 Summary of Work

This internship project developed a complete supervised pipeline for FLAIR-only brain tumor segmentation using BraTS 2020 MRI data. The work covered all important stages of a medical image segmentation workflow, including dataset selection, binary mask preparation, pre-processing, patient-level splitting, model training, threshold tuning, post-processing, quantitative evaluation, and qualitative result analysis.

The study began with an improved 2D U-Net baseline and then extended the pipeline with two attention-based variants: a self-attention U-Net and a cross-attention-gate U-Net. An ensemble prediction strategy was also examined in order to combine the strengths of the attention-based models.

8.2 Key Findings

The final selected ensemble achieved a patient-level test Dice score of 0.8667, with corresponding IoU, sensitivity, precision, and specificity values of 0.7741, 0.8678, 0.8824, and 0.9988 respectively. These results show that meaningful whole-tumor segmentation can be achieved even when only the FLAIR modality is used.

The project successfully demonstrates an end-to-end and reproducible segmentation pipeline that is appropriate for internship-level deep learning work and provides a strong base for future improvement through multimodal input, 2.5D or 3D modelling, and stronger training strategies.

8.3 Future Scope

Future work can extend this project by adding T1, T1ce, and T2 modalities, using 2.5D or 3D U-Net models, testing Dice-Tversky or focal Tversky loss, improving sampling and augmentation, evaluating external datasets, and building a simple interface for MRI upload and mask visualization.

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